



Car Self-Parking Artificial Intelligent

Introduce:

Car Self-Parking is a system performance automatically parking by car it self, which has control turn arrow automatically and distance measuring. Beside, user control brake, gear, and accelerator. Just press the bottom self-parking, the system will compare the way your park. You do not need to controls your steering wheels. The system that control over your steering wheel which would turn and lead your car to park your car in the exact spot by calculating that width and the depth of the parking lot that you choose and a user just control pedal and brake.

Summarizes the PEAS description for the car self-parking task environment.

Agent Type	Performance Measure	Environment	Actuator	Sensors
Car self-parking	Safe, Precisely-parking, Convenient	Car park, Parking lots, Parking lane	Signal display, Turn-automatically, measuring, Electric power assist steering	Cameras, Engine sensors, Sonar

Descriptions:

Performance measure:

- 1) Safe: The objectives of the system are that when a user parks a car, the system would calculate the distance between each object that is surrounding your car so that the user is safe and prevented from accident.
- 2) Precisely-parking: When a user control pedal and brake of a car, the system would calculate the distance correctly and display of each object that is surrounding your car so that the user can see on monitor inside car.
- 3) Convenient: When a user parks a car, they control only pedal and brake of a car so that car will be turn automatically for you.

Environment:

Any car must deal with a variety of parks such as car park, parking lane, parking lots.

Actuator:

- 1) Signal display: The display monitor inside car would show distance between each object that is surrounding your car front and back, a user must be able to see front and back of a car to controlling pedal and brake.
- 2) Turn-automatically: The system would control your steering wheel automatically when your switch gear to control your car go front and back.
- 3) Measuring: The system must be able to show your distance so that the system would calculate automatically between each other objects.
- 4) Electric power assist steering: the system that control over your steering wheel which would turn and lead your car to park your car in the exact spot by calculating that width and the depth of the parking lot that you choose.

Sensors:

- 1) Cameras: The cameras sends overall environment of parks to display monitor on your car.
- 2) Engine sensors: The sensor that put surrounding your car, which manage distance when you parks a car.
- 3) Sonar: The sonar will be beeping when your car close to any objects.

Summarize of task environment and their characteristics.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Car Self-Parking	partially	Single agent	Stochastic	Sequential	Dynamic	Continuous

Descriptions:

Partially observable: The agent sensors has check distances of parks environment every step if the user switch gear for driving backward then, sensors must be calculate between your car and each object on backside.

Single agent: Because, only one car interact parking by itself, so it is single agent.

Stochastic: The agent would be check every time when a user park a car in each environment and many parking environment are not the same.

Sequential: The current decision could affect all future decision, because when the agent decision to turn steering wheels in each time that has effect on the next turn in future.

Dynamic: Parking environment are continuously asking the agent to calculate distance front and back and then, the agent decision to turn steering wheels on next steps.

Continuous: When a user parks a car, they must be continuous until parking successfully.

